Leukocyte Migration

Leukocyte (white blood cell) migration is how circulating leukocytes leave the blood and reach the site of injury or infection. It occurs mainly in **post-capillary venules**, where blood flow is slowest and the endothelium is most responsive. The steps are: **margination and rolling → adhesion → transmigration (diapedesis) → chemotaxis**.

1. Margination and rolling (selectins)

- In normal flow, leukocytes travel in the centre of the vessel. At the inflamed site, **vasodilation** and increased permeability slow the blood (**stasis**), so leukocytes drift to the vessel wall. This is **margination**.
- Inflammatory mediators (**histamine, thrombin, TNF** [tumour necrosis factor], **IL-1** [interleukin-1]) activate the endothelium to express **selectins**:
    - **P-selectin**: stored in **Weibel-Palade bodies** and moved to the surface within minutes (by histamine, thrombin).
    - **E-selectin**: made by new gene expression, appearing after about 1-2 hours (induced by TNF and IL-1).
- Selectins bind **sialyl-Lewis X** on leukocyte glycoproteins (e.g., **PSGL-1**, P-selectin glycoprotein ligand 1).
- This binding is **weak and reversible**, so leukocytes **roll** along the wall. **L-selectin** on leukocytes also binds sialylated ligands on the endothelium.

2. Firm adhesion (integrins)

- **Chemokines** displayed on the endothelial surface bind receptors on the rolling leukocyte, which triggers **integrin activation** (a conformational change to high affinity).
- Endothelial cells (stimulated by TNF and IL-1) express integrin ligands:
    - **ICAM-1** (intercellular adhesion molecule 1) binds **LFA-1** (leukocyte function-associated antigen 1, CD11a/CD18) and **Mac-1** (CD11b/CD18).
    - **VCAM-1** (vascular cell adhesion molecule 1) binds **VLA-4** (very late antigen 4).
- This **strong binding** stops the leukocyte and anchors it to the endothelium.

3. Transmigration (diapedesis)

- Leukocytes squeeze **between endothelial cells** (paracellular route, the main one), moving through at the junctions.
- Key molecules: **PECAM-1** (platelet endothelial cell adhesion molecule 1, also called **CD31**) on both leukocytes and endothelial junctions, plus other junction proteins (e.g., **JAMs**, junctional adhesion molecules, and CD99).
- Leukocytes then cross the **basement membrane** by secreting **collagenases** and other proteases.

4. Chemotaxis

Once in the tissue, leukocytes move along a **chemical gradient** towards the injury.

Chemoattractants:

- **Bacterial products**: N-formylmethionine peptides
- **Complement fragments**: **C5a** (strongest), C3a
- **Leukotriene B4 (LTB4)**: from arachidonic acid
- **Chemokines**: especially **IL-8** (CXCL8, for neutrophils)
- Cytokines and other host-derived signals

Mechanism:

- Chemoattractants bind **G-protein-coupled receptors** on the leukocyte, activating signalling (e.g., **Rho GTPases**).
- This triggers **actin polymerisation** at the leading edge: the cell forms **pseudopodia** and crawls forward, with **myosin** at the rear.
- Leukocytes also become **activated** (increased phagocytosis, killing, and degranulation).

Which cells arrive when?

Time	***Main cell***	***Why***
*First **6-24 hours***	**Neutrophils**	Most abundant, respond quickly to chemokines (IL-8) and C5a, adhesion molecules appear early
24-48 hours	**Monocytes → macrophages**	Slower to arrive; neutrophils are short-lived (apoptose); monocyte-specific chemokines (MCP-1, monocyte chemoattractant protein 1)
Viral infection	**Lymphocytes** first	Specific chemokines

Allergy, parasitic	**Eosinophils**	Eotaxin

Clinical correlations: leukocyte adhesion deficiencies (LAD)

- **LAD type 1:** defect in the **CD18 β2 integrin subunit** (needed for LFA-1 and Mac-1). Neutrophils cannot adhere firmly. Features: **delayed umbilical cord separation**, recurrent bacterial infections, **no pus formation** (neutrophils cannot leave vessels), and **very high blood neutrophil count (leukocytosis)**.
- **LAD type 2:** defect in **fucose metabolism**, so **sialyl-Lewis X** is not made and rolling fails. Also causes recurrent infections, with a specific blood group phenotype (Bombay).

Therapeutic relevance: Anti-integrin antibodies (e.g., **natalizumab**, against α4 integrin, used in multiple sclerosis and Crohn's disease) block leukocyte entry into tissue, and also carry infection risks.

Summary

Step	***Key molecules (endothelium ↔ leukocyte)***	***Result***
Rolling	P-selectin, E-selectin ↔ sialyl-Lewis X / PSGL-1	Weak, reversible adhesion
Activation	Chemokines → integrin activation	Integrins shift to high affinity
Firm adhesion	ICAM-1 ↔ LFA-1, Mac-1; VCAM-1 ↔ VLA-4	Stops leukocyte
Transmigration	PECAM-1 (CD31) ↔ PECAM-1	Crosses endothelium and basement membrane
Chemotaxis	C5a, IL-8, LTB4, bacterial peptides	Moves towards injury

Memory line: selectins make leukocytes **roll**, integrins make them **stick**, PECAM-1 lets them **squeeze through**, and chemokines **guide** them.